Teaching notes

This could be a starter, plenary or revision activity.

It can be carried out using scraps of paper, exercise books, with laminated cards, or on an interactive whiteboard. There is a Magnet interactive resource that can be used with the activity.

Working in pairs, students look at the words and phrases, trying to choose four which are connected in some way. They agree on one or two sentences which include the four words and explain the connection between those words. They can then put up their hands; the class is stopped and everyone listens to the connection. If the explanation is agreed to be valid, those four words are scrubbed from the list and everyone has to start again with the remaining words and phrases.

It is up to you to decide whether to continue with the game if a combination of words does not match one of the suggested answers. You could praise students for their ingenuity but urge the class to continue to search for a group of four which matches the suggested answers; this way you will not be left with a group of mismatched words at the end. (However, your class might enjoy rising to the challenge of linking, for example, recessive, meiosis, clone and Franklin!)

Suggested answers

haemophilia	sickle cell anaemia	cystic fibrosis	colour blindness
vegetative reproduction	asexual	clones	identical
A	C	T	G
Levenne and Chargraff	Franklin	Watson	Crick
chromosome	DNA	bases	gene
homozygous	heterozygous	recessive	dominant
meiosis	mitosis	haploid	diploid

Haemophilia, **sickle cell anaemia**, **cystic fibrosis** and **colour blindness** are all examples of inherited diseases.

Vegetative reproduction is an example of **asexual** reproduction in which the offspring are all **clones** and **identical** to the parent plant.

A,**C**,**T** and **G** are the four bases found in DNA. A always pairs with T and C with G.

These people all worked on the structure of DNA. **Levene and Chargaff** described the base pairs, whilst **Franklin**, **Watson** and **Crick** worked out that the DNA molecule was a double helix.

Chromosomes contain **DNA**. Sequences of **bases** in DNA make up the **code** for different **genes**.

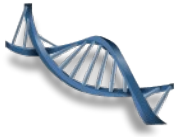
Characteristics are controlled by pairs of alleles. If the alleles in the pair are identical, the gene is **homozygous**; if they are not identical, it is known as **heterozygous**. One of the alleles will be **dominant** and the other **recessive**.

Meiosis and **mitosis** are both types of cell division. Meiosis forms **haploid** daughter cells whereas mitosis forms **diploid** daughter cells.

Of course, students may come up with their own combinations which are equally as valid.

Activity

1. Cut up the tiles.
2. Choose four tiles which are connected.
3. Write one or two sentences which include all of the four words and which explain their connection.
4. Continue until you have used all the tiles.



vegetative reproduction	A	sickle cell anaemia	Crick
homozygous	asexual	T	identical
Franklin	bases	gene	haploid
chromosome	haemophilia	Watson	colour blindness
meiosis	DNA	cystic fibrosis	mitosis
recessive	Levenne and Chargraff	G	clones
C	heterozygous	dominant	diploid

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